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PROPOSAL ON : Mekelle Fuel Tracker :
Smart Madya Platform for Fuel Management

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Abstract

Fuel distribution inefficiency has become a major socio-economic challenge in Mekelle due to persistent shortages, manual allocation systems, lack of transparency, and weak monitoring mechanisms. The current system leads to long queues, inequitable distribution, fuel hoarding, and black-market activities.

This project proposes the design and development of a **Smart Madya Platform**, a digital fuel distribution and management system that integrates government authorities, fuel stations, and vehicle owners into a unified, real-time ecosystem. The system aims to automate fuel quota allocation, improve transparency, and ensure equitable access through intelligent monitoring and control.

The proposed solution consists of three core subsystems:

- **Admin Web Dashboard** for centralized control and policy enforcement
- **Fuel Station Interface** for transaction processing and verification
- **Driver Mobile Application** for accessing fuel services

Key functionalities include:

- Digital quota allocation and tracking
- Virtual queuing system to minimize congestion
- Secure digital payment integration (e.g., Telebirr, CBE)
- Real-time notifications and fuel availability updates
- Fraud detection and reporting mechanisms
- Offline and SMS-based access for low-connectivity environments

The system is expected to significantly improve efficiency, fairness, and transparency in fuel distribution, while reducing operational inefficiencies and illegal activities.

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1. Introduction

Fuel is a critical resource that supports transportation, logistics, and economic activities. In urban centers such as Mekelle, the availability and fair distribution of fuel directly impact daily life and economic stability.

Currently, the fuel distribution system operates largely through manual processes, which suffer from:

- Inefficiency and delays
- Lack of real-time data
- Poor coordination among stakeholders
- High susceptibility to corruption and misuse

The absence of a centralized digital system makes it difficult to monitor fuel allocation, enforce quotas, and provide accurate information to users.

With the increasing demand and limited supply, there is an urgent need for a **smart, scalable, and transparent digital system**. This project introduces a **Smart Madya Platform** that leverages modern technologies such as mobile applications, cloud computing, and real-time data processing to transform the existing system.

2. Problem Statement

The existing fuel distribution system in Mekelle faces multiple operational and structural challenges:

2.1 Operational Problems

- Long queues causing time loss and traffic congestion
- Manual verification processes leading to delays
- Inefficient communication between stakeholders

2.2 Systemic Problems

- Lack of transparency in quota allocation
- Weak accountability mechanisms
- Absence of centralized data management

2.3 Security and Economic Issues

- Fuel hoarding and illegal resale (black market)
- Identity fraud and multiple registrations
- Lack of monitoring and auditing tools

2.4 Technical Limitations

- No real-time fuel availability tracking
- Limited integration with digital payment systems
- Poor accessibility in low-connectivity environments

These challenges necessitate a robust digital solution that ensures efficiency, fairness, and accountability.

3. Objectives

3.1 General Objective

To design and develop a smart, secure, and scalable digital platform that enhances fuel distribution efficiency, transparency, and fairness in Mekelle.

3.2 Specific Objectives

- To design a centralized system for managing vehicles, users, and fuel quotas
- To develop a mobile application for drivers to monitor fuel allocation and queue status
- To implement a station-level system for real-time transaction recording
- To integrate digital payment systems such as Telebirr/Chappa
- To develop a virtual queue system that reduces physical congestion
- To implement fraud detection algorithms and reporting mechanisms
- To provide real-time notifications using push notifications
- To ensure system usability in low-connectivity environments through offline support

4. Scope of the Project

4.1 In Scope

- Digital fuel quota allocation and management
- Real-time fuel tracking and reporting
- Multi-platform system (Web + Mobile)
- Role-based access control (Admin, Station Manager, Driver)
- Integration with payment and notification systems

4.2 Out of Scope / Limitations

- Full integration with national ID databases (optional simulation)
- Direct hardware integration with fuel pumps (simplified simulation)
- Deployment at national scale (prototype level implementation)
- Dependence on external APIs (payment)

5. Literature Review

Digital fuel management systems have been implemented globally to improve efficiency and transparency. These systems often incorporate:

- **IoT-based monitoring systems** for fuel tracking
- **Mobile applications** for user interaction
- **Cloud-based architectures** for centralized data storage
- **E-governance platforms** for public resource management

Key Findings from Existing Studies

- Digital queue systems significantly reduce waiting time
- Mobile payments increase transparency and traceability
- Centralized systems improve monitoring and reporting
- Real-time data enhances decision-making

Research Gap

Most existing systems are designed for high-connectivity environments and lack:

- Offline functionality
- Adaptability to developing regions

This project addresses these gaps by designing a system suitable for **low-resource and low-connectivity environments**.

6. Methodology

The project will follow the **Agile Software Development Methodology**, which emphasizes iterative development, flexibility, and continuous feedback.

6.1 Agile Phases

1. Requirement Analysis
2. System Design
3. Development (Iterative Sprints)
4. Testing and Validation
5. Deployment and Evaluation

6.2 Justification for Agile

- Supports changing requirements
- Encourages continuous stakeholder feedback
- Enables incremental delivery and Reduces development risk

7. Proposed System Architecture and Design

7.1 System Architecture

The system follows a **3-Tier Architecture**:

1. Presentation Layer

- Mobile Application (Drivers)
- Web Dashboard (Admin & Station Managers)

2. Application Layer

- Backend APIs (Business Logic)
- Authentication & Authorization Services

3. Data Layer

- Centralized Database Server
- Data Storage and Retrieval

7.2 Use Case Design

Actors:

- Admin (Government Authority)
- Station Manager
- Driver

Major Use Cases:

- Manage fuel quotas
- Register vehicles and users
- Track fuel availability
- Join virtual queue
- Process payments
- Generate reports

7.3 Sequence Flow

1. Driver requests fuel via mobile app
2. System verifies quota
3. Driver joins virtual queue
4. Station confirms availability
5. Payment is processed
6. Fuel is dispensed
7. Transaction is recorded

7.4 Database Design

Main Entities:

- Users (Admin, Driver, Staff)
- Vehicles
- Fuel Quotas
- Transactions
- Stations
- Payments

Relationships:

- One user → many vehicles
- One vehicle → one quota
- One station → many transactions

8. Tools and Technologies

Frontend

- Next.js (Web Application)
- React Native (Mobile App)

Backend

- Nestjs

Database

- PostgreSQL

Other Technologies

- RESTful APIs
- Firebase Cloud Messaging (Notifications)
- Payment APIs (Telebirr/chappa)
- Git & GitHub (Version Control)

9. Implementation Plan / Work Schedule

Phase	Activity	Duration
Phase 1	Requirement Analysis	1 Weeks
Phase 2	System Design (UML, ERD)	
Phase 3	Development (Frontend & Backend)	3 Weeks
Phase 4	Testing & Debugging	1Weeks
Phase 5	Deployment & Documentation	

10. Expected Results

The system is expected to:

- Reduce long queues and congestion at fuel stations
- Ensure fair and transparent fuel distribution
- Improve accountability and monitoring
- Minimize fuel fraud and illegal activities
- Provide real-time fuel information
- Enhance user experience through mobile access
- Support decision-making through analytics and reports

References

- Research papers on Smart Fuel Management Systems
- Agile Software Development Literature
- Mobile Application Development Guides
- REST API Design Standards
- Cloud Computing and Database Design Resources